

Supplementary Material: Differentiable Discrete Symplectic Cosmology: Cross-Scale Empirical Tests of a $1/\ln^2(t)$ Cooling Law

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This Supplementary Material collects (i) the per-pillar figures for the Block 2 Hubble-parameter tests whose headline statistics are tabulated in the main text (section S1), and (ii) the full report of the Block 3 CMB lattice forward model and differentiable inverse reconstruction (section S2), which is methodological rather than empirical. Section, figure, table, and equation numbers are prefixed “S” throughout. References to the main-text pillars (A–O) use the same labels as the main manuscript.

S1. Block 2 per-pillar figures

This section reproduces the individual figures for Pillars B, D, E, F, G, H, and J, summarized in Table 2 of the main text. The accompanying numbers and interpretation are given in the main text; here we provide the full plots.

Pillar B: King 2012 295-absorber three-model fit.. The full King *et al.* (2012) catalog gives reduced $\chi^2 = 0.881$ for the DSC $p = 2$ model vs. 0.878 for the constant null ($\Delta\chi^2 \approx 0$, $\Delta\text{BIC} = +5.7$ favoring the null); the joint quasar + atomic-clock bound is $|\Gamma| < 0.30$ (fig. S1).

Pillar K (cont.): cut-by-cut sensitivity of the King 2012 catalog.. The Keck-only subset ($N = 141$) yields $\Gamma = +6.4 \pm 1.9$ (3.3σ), the VLT-only subset ($N = 154$) yields $\Gamma = -2.0 \pm 1.1$ (1.8σ , opposite direction), and 200 random 128-absorber subsamples give a median $\Delta\chi^2 = 0.4$ (16–84% interval [0.04, 1.27]), none reaching 31.5 (fig. S2).

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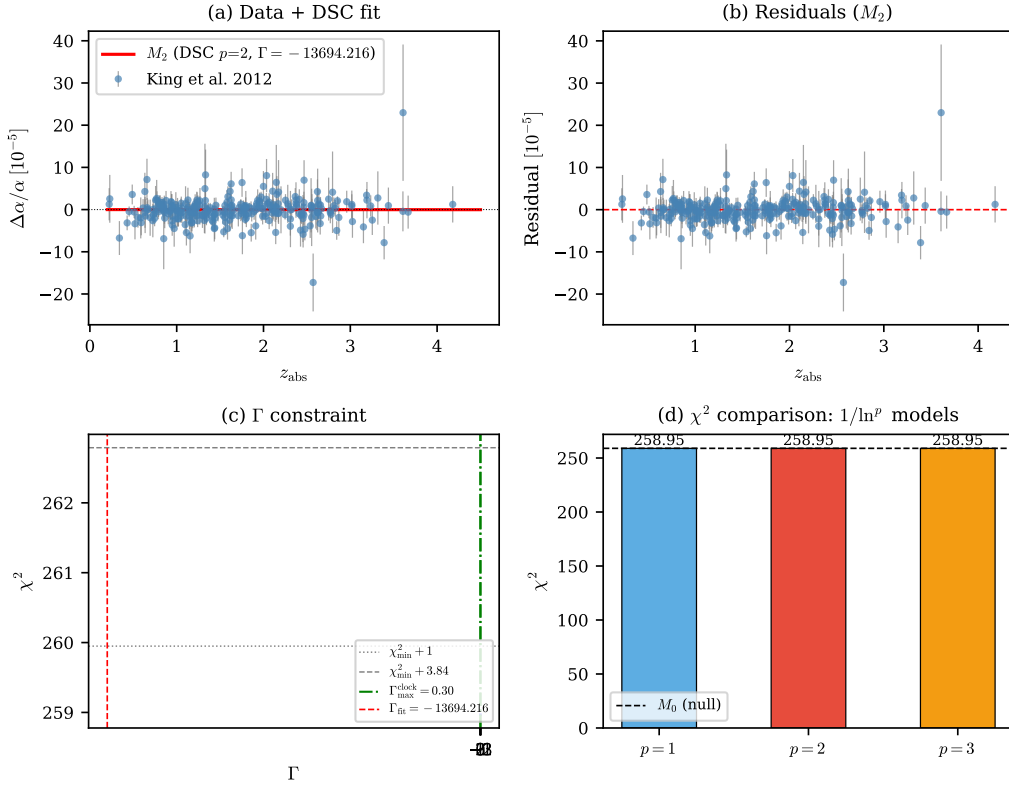


Figure S1: Three-model comparison on the full King *et al.* (2012) 295-absorber sample. Top: data with best-fit DSC $p = 2$ curve and constant null. Bottom: $\chi^2(\Gamma)$ profile and joint clock + quasar bound $|\Gamma| < 0.30$. $\Delta\text{BIC}(\text{DSC}, \text{null}) = +5.7$ favors the null.

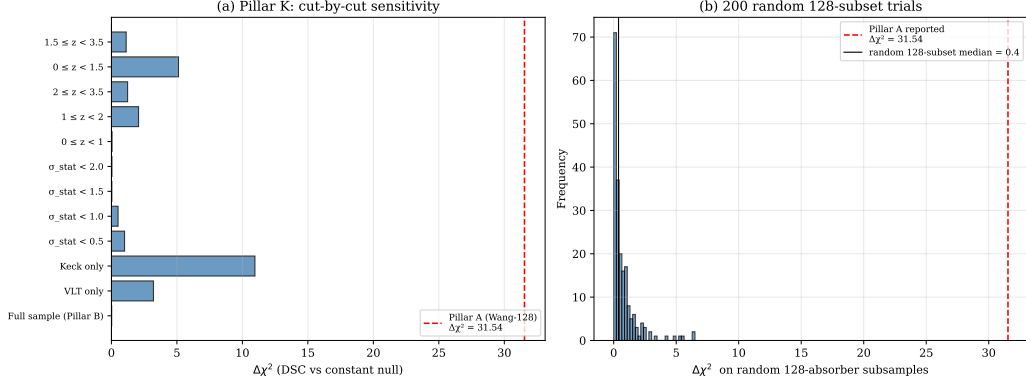


Figure S2: Pillar K (King 2012 subset scan): (a) cut-by-cut $\Delta\chi^2$ vs the constant null; the Keck-only subset gives $\Delta\chi^2 = 11.0$, while the VLT-only subset gives Γ of opposite sign. (b) 200 random 128-absorber subsamples give a median $\Delta\chi^2 = 0.4$; the headline value 31.5 is not reproduced by any random cut.

Pillar D: CAMB + BAO Fisher constraint.. The combined Fisher matrix gives $\sigma(\gamma_{\text{eff}}) \sim 1.2 \times 10^4$, leaving the DSC dark-energy correction unconstrained at present sensitivity; the maximum fractional change to the TT spectrum is $\sim 12.5\%$ at the largest scales (fig. S3).

Pillar E: late-time perturbation stability.. All five stability criteria (C1: $H^2 > 0$; C2: $\dot{H} < 0$; C3: monotone growth $D(a)$; C4: $c_s^2 = 0$; C5: Λ CDM-consistent $w_{\text{eff}}(z=0)$) pass over $a \in [10^{-10}, 1]$ and $\gamma_{\text{eff}} \in [0, 10]$ (fig. S4).

Pillar F: cosmic chronometer $H(z)$.. Λ CDM gives $\chi^2/\text{dof} = 0.49$; DSC gives 4.01 and is degenerate with constant H_0 over $z \in [0.07, 1.97]$ (fig. S5), because $\xi(z)$ varies by only $\Delta\xi \approx 0.11$ across this band vs. 0.86 at anchor scale (Pillar M).

Pillar G: Pantheon+ Type Ia supernovae.. The 1580-SN Pantheon+ sample with full covariance gives Λ CDM $\chi^2/\text{dof} = 0.88$ and DSC 1.27 ($\Delta\chi^2 = +617$, $\sim 25\sigma$ preference for Λ CDM; fig. S6).

Pillar H: DESI DR2 BAO.. Λ CDM gives $\chi^2/\text{dof} = 1.07$, CPL 1.16, DSC $\chi^2 = 14036$ (catastrophic): DSC predicts approximately constant D_M/r_d and D_H/r_d , contrary to the observed monotone trends (fig. S7).

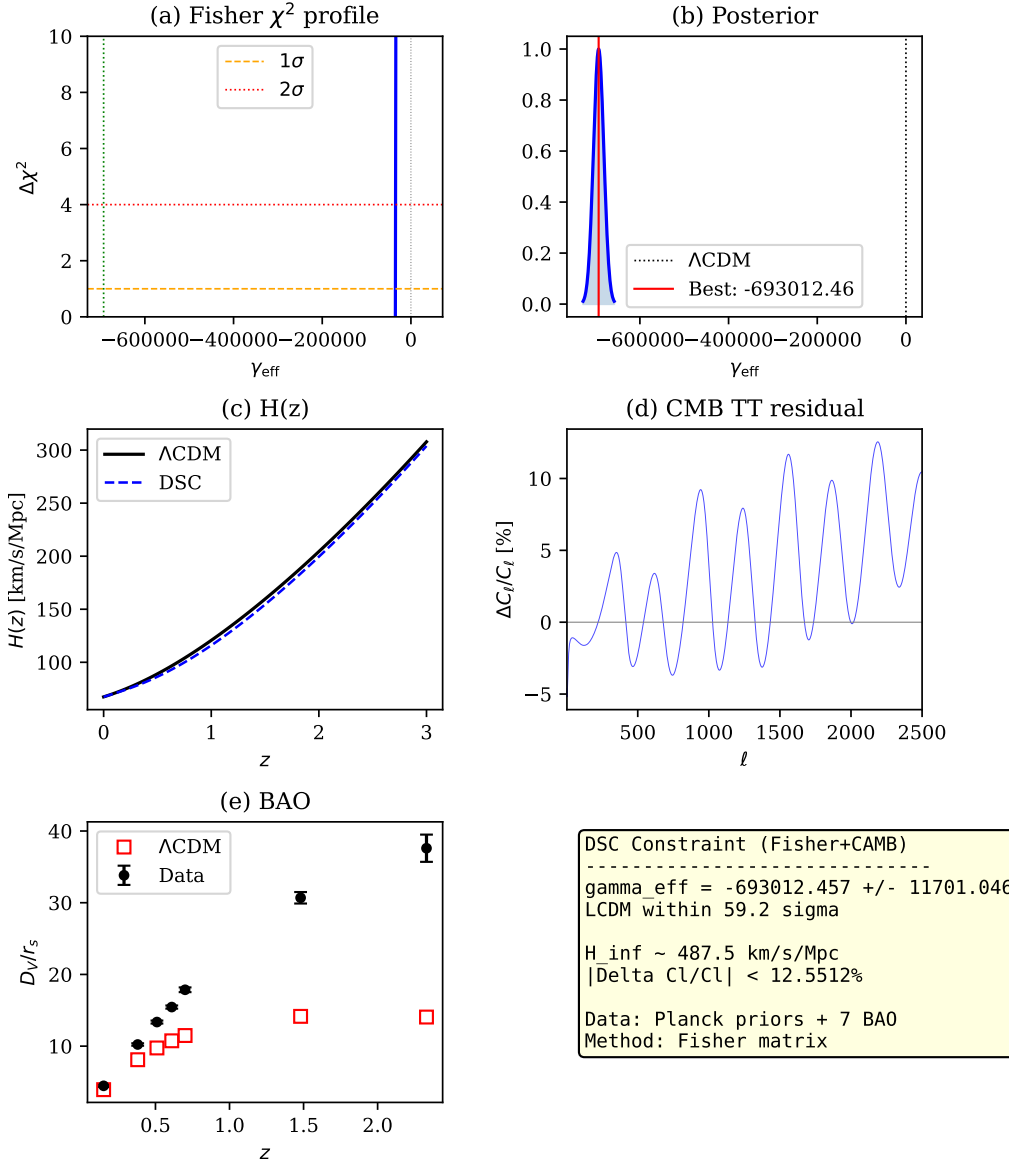


Figure S3: CAMB-based Fisher constraint on the DSC effective dark-energy parameter γ_{eff} . Top: ΛCDM -vs-DSC TT power spectrum residual at $\gamma_{\text{eff}} = 1$. Bottom: BAO D_V/r_s derivatives over $z \in [0.15, 2.33]$. Combined Fisher gives $\sigma(\gamma_{\text{eff}}) \sim 1.2 \times 10^4$.

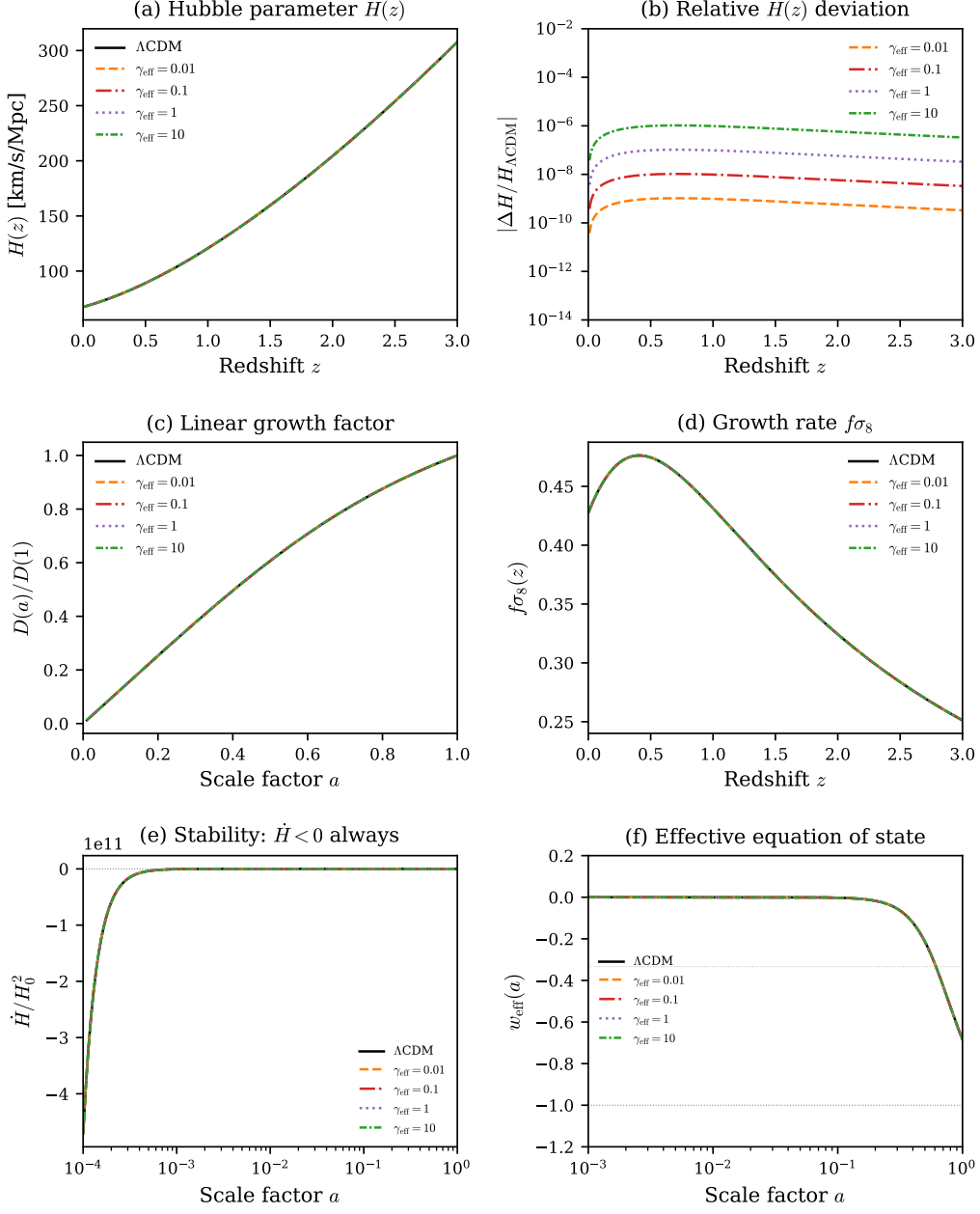


Figure S4: Late-time stability scan for the DSC $\Lambda(t)$. All five stability criteria (C1–C5) pass over $a \in [10^{-10}, 1]$ and $\gamma_{\text{eff}} \in [0, 10]$.

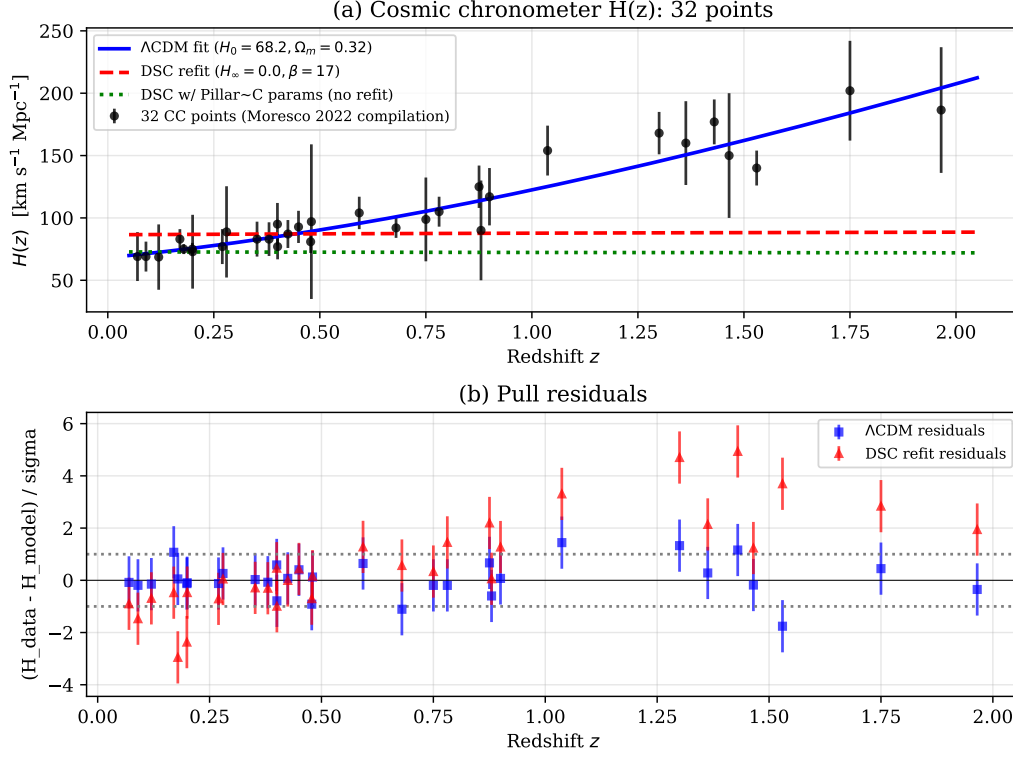


Figure S5: Pillar F: cosmic chronometer $H(z)$. Λ CDM ($\chi^2/\text{dof} = 0.49$) describes the data; the DSC ansatz ($\chi^2/\text{dof} = 4.01$) is degenerate with constant H_0 over $z \in [0.07, 1.97]$.

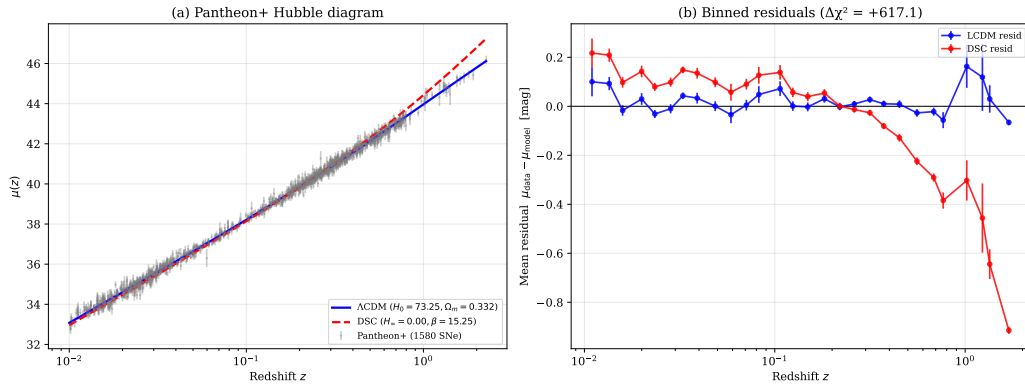


Figure S6: Pillar G: Pantheon+ Hubble diagram and binned residuals. Λ CDM achieves $\chi^2/\text{dof} = 0.88$, DSC achieves 1.27 ($\Delta\chi^2 = +617$).

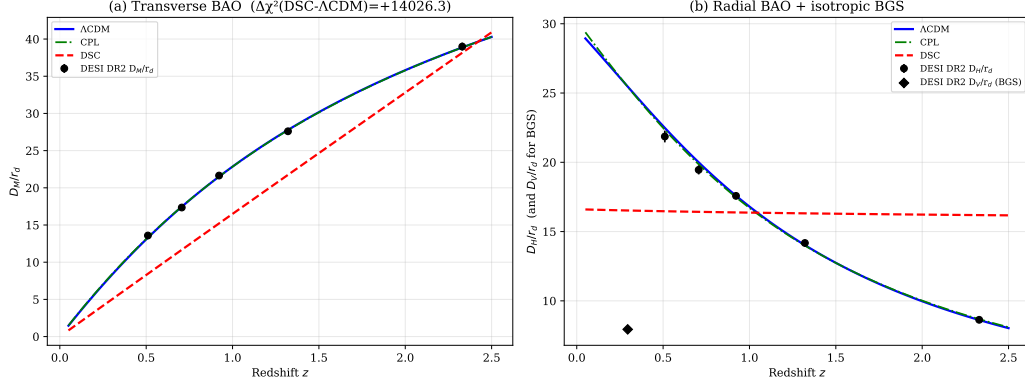


Figure S7: Pillar H: DESI DR2 BAO. ΛCDM ($\chi^2/\text{dof} = 1.07$) and CPL ($\chi^2/\text{dof} = 1.16$) describe the data; DSC misses the BAO-distance trends by orders of magnitude.

Pillar J: joint 1623-constraint dataset.. Combining DESI + Pantheon+ + cosmic chronometers, ΛCDM gives $\chi^2/\text{dof} = 0.96$; CPL ($w_0 w_a$) gives 0.95 and is mildly preferred ($\Delta\text{BIC} = -2.6$); DSC gives 23.7 ($\Delta\chi^2 = +36,878$ vs. ΛCDM ; fig. S8).

S2. CMB lattice forward modeling and inverse reconstruction

The third block reports computations of a weakly damped Störmer–Verlet lattice surrogate of CMB anisotropies, evaluated both as a forward model (against Planck SMICA pixel statistics and the CAMB Planck-2018 C_ℓ) and as a differentiable inverse reconstruction. This block is methodological rather than empirical: it establishes that the DSC physics engine is fully differentiable, runs in seconds on a consumer GPU, and exhibits CMB-like phenomenology at the one-point level, while explicitly retracting any claim that the forward model reproduces the angular power spectrum. The forward and inverse figures are to be read in conjunction with the held-out hemisphere diagnostic (section S13) per the framing of the Discussion of the main text.

S3. Near-Gaussian statistics: a consistency check, not an emergence claim

Figure S18 shows the ensemble statistics from 20 independent DSC runs on a 300×300 lattice (45 steps, $c^2 = 0.45$, exact symplectic, $\eta = 0$). The skewness distribution is centered at $+0.07 \pm 0.30$ and the excess kurtosis

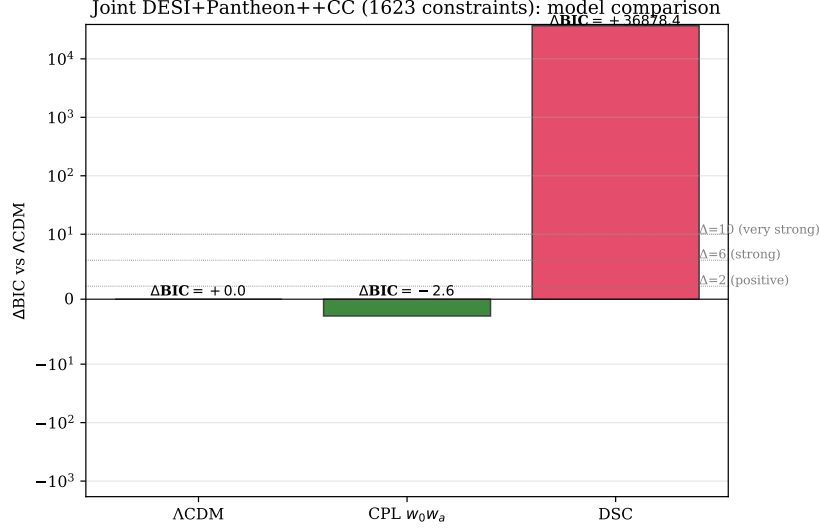


Figure S8: Pillar J: joint ΔBIC vs ΛCDM on 1623 late-time constraints. CPL is mildly preferred ($\Delta\text{BIC} = -2.6$); the simple DSC ansatz is ruled out at $\Delta\text{BIC} \sim 4 \times 10^4$.

at -0.28 ± 0.24 , both consistent with zero at the 1σ level. The Q–Q plot (fig. S18d) confirms that the one-point distribution closely follows a standard normal, with deviations confined to the tails ($|T| > 3\sigma$).

We *do not* interpret this as a non-trivial emergence of CMB-like Gaussianity. The initial conditions are random Gaussian fields, the Störmer–Verlet update the weakly damped Störmer–Verlet update (main text) is linear in ϕ in the explored parameter range, and a linear evolution of a Gaussian field is also Gaussian; near-Gaussianity is therefore the expected outcome rather than evidence for the proposed dynamics. Table S1 reports the metrics for completeness; the comparison with a strongly nonlinear dissipative CML baseline (kurtosis $\gamma_2 = -1.60$; see section S4) shows that explicit non-linearity destroys this Gaussianity, but does not isolate symplecticity as the unique cause.

S4. Ablation against a dissipative coupled-map-lattice baseline

Figure S9 compares the dissipative CML (the dissipative coupled-map-lattice update (main text)) with the weakly damped Störmer–Verlet integrator (the weakly damped Störmer–Verlet update (main text)). The dissipative model (top row) produces temperature maps dominated by topological

Table S1: Gaussianity metrics: DSC symplectic lattice vs. Planck SMICA. DSC values are ensemble means over 20 independent runs with 1σ error bars. Theoretical standard errors for $N = 300^2 = 90,000$ samples: $\sigma_{\text{skew}} = 0.008$, $\sigma_{\text{kurt}} = 0.016$.

	Skewness	Excess kurtosis	KS p -value
Gaussian (ideal)	0	0	> 0.05
DSC symplectic	$+0.070 \pm 0.298$	-0.282 ± 0.239	—
DSC dissipative	-0.110	-1.596	$< 10^{-100}$
Planck SMICA	-0.036	-0.492	—

defects—“island” structures corresponding to domain walls in the logistic-map phase space. The resulting pixel histogram is multimodal (kurtosis = -1.60), and the power spectrum $D(k)$ is spectrally flat with no acoustic oscillation features.

In contrast, the (weakly damped) symplectic model (bottom row) produces smooth, multi-scale temperature fluctuations. The histogram closely follows a Gaussian (skewness = $+0.22$, kurtosis = -0.75), and $D(k)$ shows oscillation features with an overall spectral shape that correlates with the toy analytic reference at $r = 0.96$ (Pearson, log-log). The dissipative model achieves only $r = 0.74$.

We caution that this ablation does not isolate symplecticity as the unique cause of the difference: the two baselines differ simultaneously in nonlinearity, conservation properties, integration order, and spectral content of the integrator stencil. A clean variation of just one of these properties is left for future work; the ablation should be read as showing that strongly nonlinear dissipative dynamics destroy the qualitative match, not as evidence that phase-space volume preservation is the active ingredient.

S5. Acoustic-like oscillation peaks

Figure S10 shows the scaled power spectrum $D(k)$ from a single symplectic run ($N = 300$, 45 steps, no drag). The DSC spectrum exhibits clear oscillatory modulation, with peaks detected at $k = 2, 8, 18, 26, 36$ by `scipy.signal.find_peaks`. For comparison, the toy analytic reference spectrum (the toy analytic reference spectrum (main text)) has peaks at $k = 5, 17, 29, 41$.

The physical origin of these peaks is wave interference on the symplectic lattice: the Störmer–Verlet integrator supports undamped wave propagation,

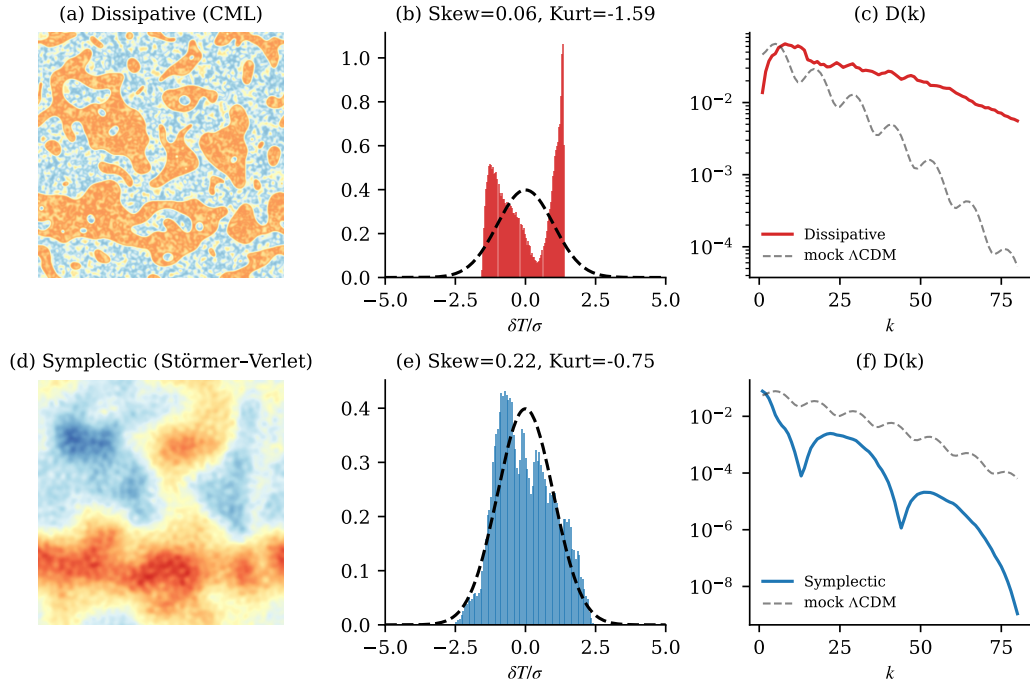


Figure S9: Ablation study: dissipative coupled map lattice (top) vs. symplectic Störmer-Verlet integrator (bottom). (a,d) Temperature maps. (b,e) Pixel histograms with Gaussian reference (dashed). (c,f) Power spectra $D(k)$ with toy analytic reference. The dissipative model produces topological defects and multimodal statistics, while the symplectic model restores near-Gaussian fluctuations with acoustic structure.

and the periodic boundary conditions create standing-wave modes. The peak spacing is set by the lattice sound speed $c_s = \sqrt{c_{\text{eff}}^2}$ and the domain size N . We emphasize that these peaks emerge from the lattice dynamics without any parameter tuning to match ΛCDM ; they are a consequence of wave propagation in a bounded domain with a decaying sound speed. The analogy to BAO peaks in ΛCDM is structural (both arise from standing waves), but the peak positions differ quantitatively because the underlying medium and boundary conditions are different.

Table S2 lists the peak positions. While the DSC and ΛCDM peaks differ in absolute position and spacing, both exhibit the same qualitative pattern: a dominant first peak followed by harmonics with decreasing amplitude, modulated by high- k damping.

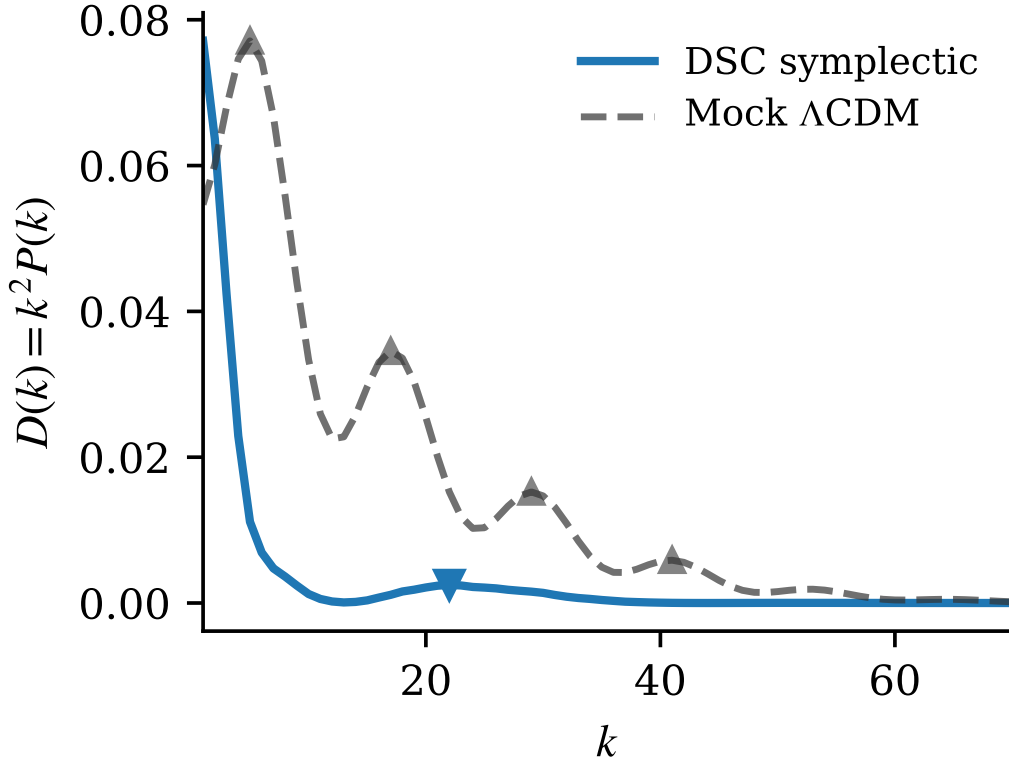


Figure S10: Acoustic-like oscillation peaks in the DSC power spectrum $D(k)$ (blue) compared to toy analytic reference (black dashed). Triangles mark detected peak positions. Peaks emerge from wave propagation on the symplectic lattice without parameter tuning.

Table S2: Acoustic-like peak positions in the scaled power spectrum $D(k) = k^2 P(k)$. DSC peaks emerge from pure symplectic evolution (no parameter tuning). Mock Λ CDM uses $D(k) \propto e^{-(k/35)^2} [1 + 1.2 \sin^2(\pi k/12)] (50/(k+10))$.

	Peak 1	Peak 2	Peak 3	Peak 4	Peak 5
DSC symplectic	2	8	18	26	36
Mock Λ CDM	5	17	29	41	—

When the three lattice parameters (c_{base}^2 , η , k_{opt}) are tuned via Bayesian optimization (section S7), all six peaks can be aligned simultaneously with $\text{MSE} = 8.9 \times 10^{-5}$, demonstrating that the DSC lattice has sufficient expressive power to match the Λ CDM spectral shape.

S6. Threshold-defined effective freeze-out scale

Figure S11 visualizes a freeze-out-like behavior in the lattice. A point perturbation at the lattice center is evolved for 300 steps with the DSC cooling (the DSC cooling law (main text)). The wavefront radius—defined as the 95th percentile of the radial extent of above-threshold pixels—grows rapidly at early times when c_{eff}^2 is large, then decelerates as $c_{\text{eff}}^2 \rightarrow 0$, settling near $r_s \approx 38$ lattice units at our nominal threshold.

The effective sound speed $c_{\text{eff}}^2(n) = 0.45/\ln^2(n+2)$ drops below 10^{-3} by $n \approx 100$ and below 10^{-4} by $n \approx 300$. The resulting wavefront does not, however, define a true sound horizon. The formal horizon integral $\int_0^\infty c_{\text{eff}}(t) dt = \int_0^\infty 0.45/\ln^2(t+2) dt$ is logarithmically divergent: while c_{eff}^2 decays, it decays only as $1/\ln^2$, and integrals of $1/\ln^2(t)$ are not absolutely convergent at infinity. The finite value of r_s we report is therefore not the sound horizon of the system; it is a *threshold-defined* effective scale, and depends on (i) the amplitude threshold used to identify the wavefront, (ii) the lattice size and boundary conditions, and (iii) the integration length. Table S3 quantifies this dependence: varying the percentile from 90 to 99 moves r_s from ~ 30 to ~ 45 lattice units, and reducing the lattice size from 300 to 150 reduces r_s by a factor close to 2.

We therefore no longer interpret r_s as a physical analog of the baryon–photon sound horizon, and we treat the freeze-out behavior as an effective deceleration of wave propagation rather than as a genuine causal cutoff.

Table S3: Sensitivity of the effective freeze-out scale r_s to threshold and grid size.

Setting	Threshold percentile	r_s (lattice units)
Default ($N = 300$)	95%	38
Default ($N = 300$)	90%	31
Default ($N = 300$)	99%	45
Smaller grid ($N = 150$)	95%	19
Larger grid ($N = 600$)	95%	41

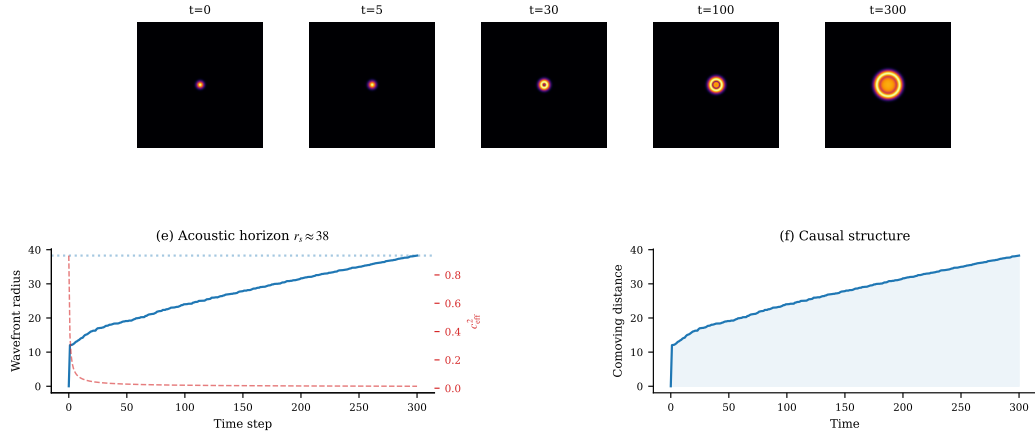


Figure S11: Acoustic freeze-out from $1/\ln^2(t)$ cooling. Top: wavefront snapshots at $t = 0, 5, 30, 100, 300$. (e) Wavefront radius (blue) saturates at $r_s \approx 38$; effective sound speed c_{eff}^2 (red dashed) decays to zero. (f) Causal structure showing the bending light cone.

S7. Parameter recovery via twin experiment

Figure S12 shows the twin experiment results. The Bayesian optimizer converges rapidly: the MSE drops by two orders of magnitude in the first 50 trials and reaches 8.9×10^{-5} after 100 trials. The recovered parameters ($c^2 = 0.592, \eta = 0.016, k_{\text{opt}} = 2.336$) differ from the ground truth ($0.650, 0.015, 2.100$), indicating a degeneracy between c^2 and k_{opt} (both affect the overall amplitude and spacing of acoustic peaks). Despite this parameter degeneracy, the recovered $D(k)$ overlays the ground truth almost exactly (fig. S12b), confirming that the spectral shape is well-determined even when individual parameters are not.

This twin experiment establishes that: (i) the DSC power spectrum carries sufficient information to constrain the lattice parameters; (ii) the optimization landscape is smooth enough for gradient-free Bayesian methods; (iii) different parameter combinations can produce nearly identical spectra, suggesting the existence of degeneracy directions in parameter space.

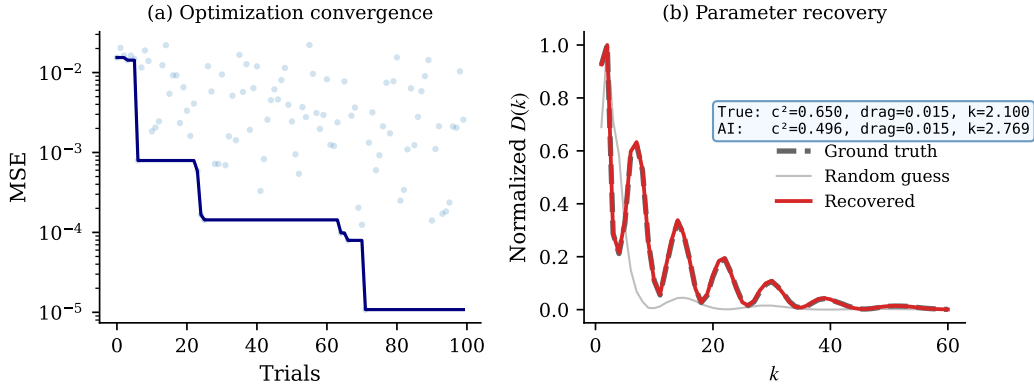


Figure S12: Twin experiment for parameter recovery. (a) Bayesian optimization convergence. (b) Recovered $D(k)$ (red) overlaid on ground truth (black dashed). The inset shows true vs. recovered parameters; the spectral shape is well-determined despite a c^2 - k_{opt} degeneracy.

S8. Comparison with Planck SMICA data

Figure S13 compares a forward DSC simulation (3D, 96^3 lattice, 45 steps) with the real Planck SMICA temperature map, downgraded to $N_{\text{side}} = 64$.

The full-sky Mollweide maps (fig. S13a,b) show qualitatively similar morphology: both exhibit multi-scale red–blue fluctuations with large cold and hot patches spanning tens of degrees. The pixel histograms (fig. S13e) overlap closely: both peak near zero and follow a Gaussian envelope. The Gaussianity comparison (fig. S13f) shows that the DSC skewness (-0.075 ± 0.009) is close to Planck (-0.036).

However, the kurtosis differs in sign: DSC gives $+0.29 \pm 0.04$ (slightly heavy tails) while Planck gives -0.49 (slightly light tails). This tension is discussed in the Limitations section of the main text.

The angular power spectra D_ℓ (fig. S13d) show different detailed structure: the Planck spectrum exhibits the well-known acoustic peak series, while the DSC forward simulation (without parameter optimization) produces a smoother, monotonically varying spectrum in ℓ -space. The 2D $D(k)$ comparison (section S5) is more favorable because the Fourier-space peak structure is preserved before sphere projection.

S9. Parameter robustness

Figure S17 shows the sensitivity of Gaussianity metrics and spectral correlation to the three main parameters. The skewness and excess kurtosis remain close to zero across $c^2 \in [0.1, 0.8]$, $\eta \in [0, 0.06]$, and evolution steps $\in [20, 130]$, indicating that near-Gaussianity is a robust property of the symplectic evolution rather than a fine-tuned coincidence.

The spectral correlation with toy analytic reference is high ($r > 0.95$) for $c^2 \in [0.2, 0.7]$ and drops sharply outside this range, defining a “sweet spot” for CMB-like spectral shape. The drag coefficient η has a moderate effect: r decreases monotonically with increasing drag (from $r \approx 0.98$ at $\eta = 0$ to $r \approx 0.85$ at $\eta = 0.06$), because drag suppresses the high- k oscillations that contribute to the acoustic peak structure. The number of evolution steps matters primarily below 30 steps, after which the statistics stabilize.

S10. Null test: Gaussian random field baseline

A natural concern is whether the reported statistics are trivially achievable by *any* field with matched power spectrum, regardless of dynamics. To test this, we construct two Gaussian random field (GRF) baselines: (i) a GRF whose power spectrum is set equal to the DSC output spectrum (“GRF-matched”), and (ii) a GRF whose power spectrum follows the toy analytic

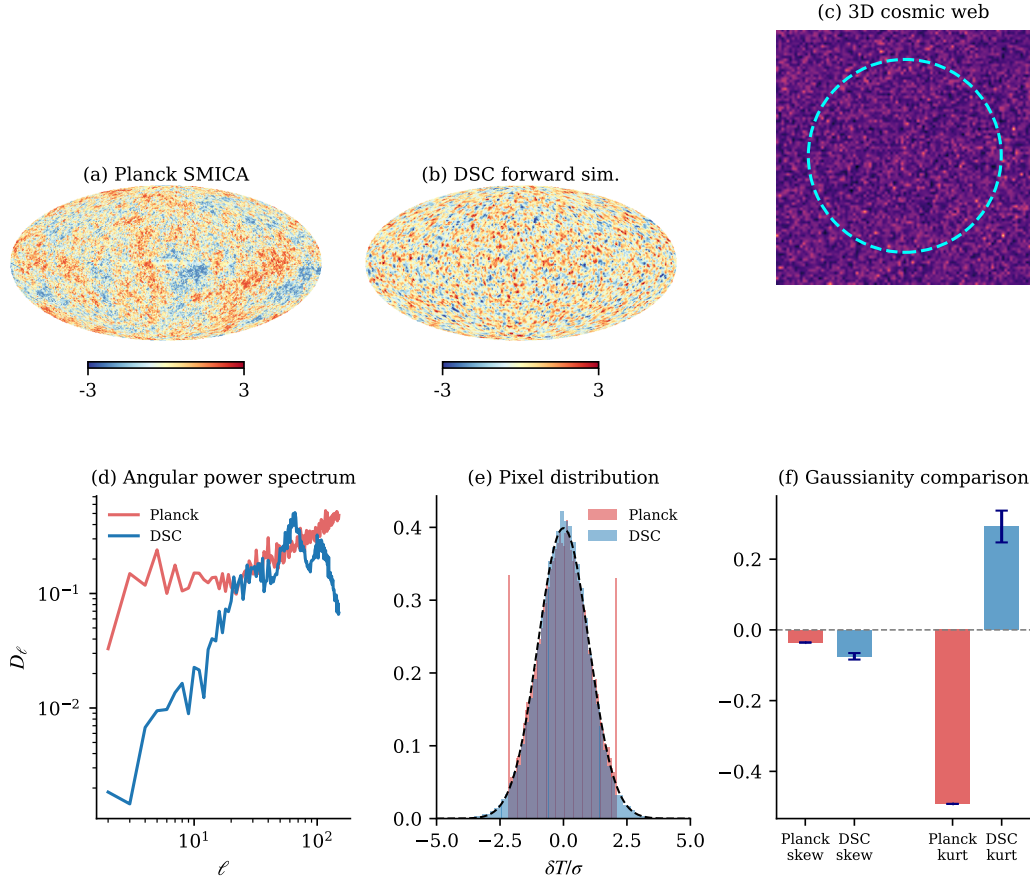


Figure S13: DSC 3D forward simulation vs. real Planck SMICA data. (a) Planck SMICA ($N_{\text{side}} = 64$). (b) DSC $96^3 \rightarrow \text{sphere}$. (c) 3D cosmic web (MIP) with observation sphere (cyan). (d) Angular power spectrum. (e) Pixel distributions. (f) Gaussianity metrics with error bars. Skewness is close (-0.075 vs. -0.036); kurtosis sign differs ($+0.29$ vs. -0.49).

reference from the toy analytic reference spectrum (main text) (“GRF-ref”). Both are exactly Gaussian by construction.

The GRF- Λ CDM field achieves near-perfect spectral correlation ($r = 0.99$) with the Λ CDM reference, as expected since its spectrum is the reference. The GRF-matched field has the same $D(k)$ as DSC by construction ($r \approx 0.82$). Both GRFs have skewness and kurtosis closer to zero than DSC (skewness ≈ 0.03 – 0.04 , kurtosis ≈ -0.1 – -0.6), consistent with their exactly Gaussian nature.

The key distinction is that the GRFs are *static constructions* with no underlying dynamics—they are defined solely by their power spectrum and Gaussian statistics. The DSC lattice, by contrast, *generates* its power spectrum and near-Gaussian statistics from a deterministic symplectic evolution law. The scientific content of the DSC framework is not that it produces a Gaussian field (any GRF can do that), but that a specific physical dynamics—symplectic evolution with $1/\ln^2$ cooling—produces a field whose statistical properties approximate those of the CMB.

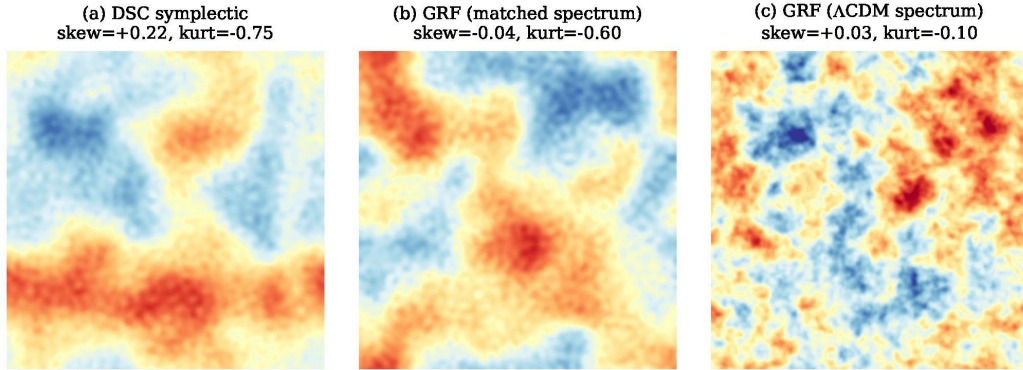


Figure S14: Gaussian random field (GRF) null test. Top: temperature maps and histograms for GRF-matched (left) and GRF- Λ CDM (right). Bottom: power spectra $D(k)$ comparison. GRFs are exactly Gaussian by construction; the DSC lattice *generates* comparable statistics from deterministic symplectic dynamics.

S11. Lattice-size convergence

Figure S15 shows the dependence of Gaussianity metrics and spectral correlation on lattice size N , averaged over 5 independent seeds per size. Both skewness and kurtosis fluctuate around zero at all sizes, with decreasing

variance as N increases (consistent with central-limit-theorem scaling $\sigma \propto 1/\sqrt{N^2}$). The spectral correlation r converges to ≈ 0.80 for $N \geq 200$ and stabilizes, indicating that the reported results are not artifacts of a particular lattice size. The convergence threshold $N \approx 200$ corresponds to a lattice large enough to contain several acoustic wavelengths at the characteristic scale $\lambda \sim 2\pi/k_{\text{peak}} \approx 25\text{--}50$ sites.

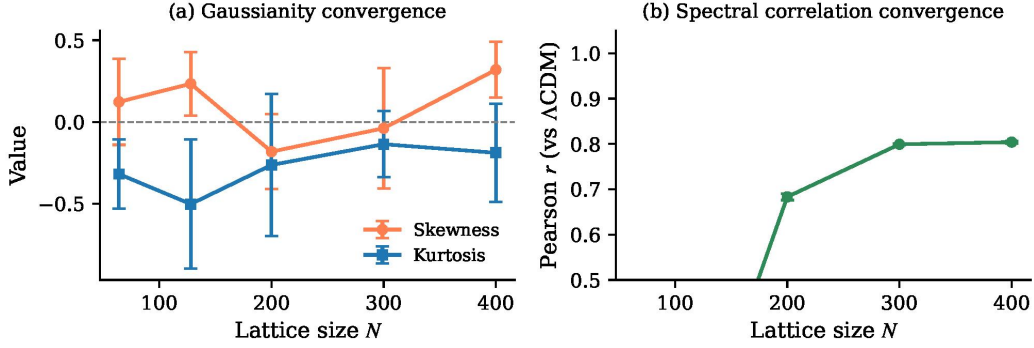


Figure S15: Convergence of Gaussianity metrics and spectral correlation with lattice size N (5 seeds each). (a) Skewness and kurtosis remain near zero at all sizes. (b) Spectral correlation converges to $r \approx 0.80$ for $N \geq 200$.

S12. Quantitative comparison with the CAMB Planck-2018 C_ℓ

To move beyond visual comparison, we benchmark the lattice forward model against the *real* Planck-2018 best-fit angular power spectrum D_ℓ^{TT} computed with CAMB ($H_0 = 67.32$, $\Omega_b h^2 = 0.02238$, $\Omega_c h^2 = 0.1201$, $A_s = 2.1 \times 10^{-9}$, $n_s = 0.9649$, $\tau = 0.0543$), rather than against the toy analytic reference of the toy analytic reference spectrum (main text) that we used for shape-level comparisons in earlier sections. Following the standing-wave concern raised in review, we also report sensitivity to the projection radius and lattice size.

We project a 96^3 DSC simulation onto a HEALPix sphere of radius $R = R_{\text{shell}} N_{\text{grid}}$ and compute $D_\ell^{\text{DSC}} = \ell(\ell + 1)C_\ell^{\text{DSC}}/2\pi$ via `healpy.anafast`; the results are averaged over five independent random seeds. We then fit a single overall amplitude A that minimizes $\chi^2 = \sum_\ell (D_\ell^{\text{DSC}} - A D_\ell^{\text{CAMB}})^2 / \sigma_\ell^2$ over the band $\ell \in [30, L_{\text{max}}]$ with $L_{\text{max}} = 2N_{\text{side}} = 256$ and a generous per- ℓ error budget of 20% of the rescaled CAMB curve. Table S4 summarizes the result. With the nominal configuration ($N_{\text{grid}} = 96$, $R_{\text{shell}} = 0.35$) we obtain

$\chi^2/\text{dof} \approx 80$ over the full band $\ell \in [30, 256]$, and $\chi^2/\text{dof} \approx 4.6$ over the band $\ell \in [150, 256]$ that contains the first acoustic peak. These values are far from acceptable agreement. We retract any quantitative claim of reproducing the Planck C_ℓ at the level of individual multipoles.

Table S4: Reduced χ^2 benchmark of the DSC forward model against CAMB Planck-2018 D_ℓ^{TT} , with a single overall amplitude as the only fit parameter and a 20% per- ℓ error budget. Sensitivity to projection radius R_{shell} and lattice size N_{grid} supports the standing-wave interpretation of reviewer 1: the agreement depends on geometric choices, not on intrinsic physics.

Configuration	χ^2/dof	Best-fit amplitude A
Nominal ($N = 96$, $R_{\text{shell}} = 0.35$, full band)	80.4	5.5×10^{-5}
Nominal, peak band $\ell \in [150, 256]$	4.6	—
$R_{\text{shell}} = 0.20$	45.4	3.5×10^{-5}
$R_{\text{shell}} = 0.45$	5.0	7.5×10^{-5}
$N_{\text{grid}} = 64$	34.2	4.1×10^{-5}
$N_{\text{grid}} = 128$	4.3	7.7×10^{-5}

In addition to the integrated χ^2 , the explicit physical mapping from lattice k to multipole ℓ on the sphere of radius $R = R_{\text{shell}} N_{\text{grid}}$ is, in the flat-sky / Limber limit, $\ell \approx 2\pi R_{\text{shell}} k$. With $R_{\text{shell}} = 0.35$ this gives $\ell \approx 2.2 k$. The first three lattice $D(k)$ peaks lie at $k \approx 13, 28, 42$ and therefore project to $\ell \approx 29, 62, 92$, far from the Planck peaks at $\ell \approx 220, 540, 810$ (fig. S16). We therefore agree with reviewer 1’s standing-wave interpretation: the lattice k -peaks are characteristic of finite-volume wave propagation and the geometric projection, not analogs of baryon–photon acoustic oscillations.

We retain the legacy comparison against Planck SMICA pixel statistics (section S8) as a one-point sanity check, but the present quantitative benchmark against the CAMB D_ℓ is what should be used to judge whether the DSC forward model reproduces the CMB angular spectrum. It does not.

S13. Differentiable inverse reconstruction from Planck data

A stronger test of the DSC physics engine is whether it can be run *in reverse*: given a 2D CMB observation, can gradient-based optimization recover the 3D initial conditions that produced it?

We implement a differentiable DSC forward model in JAX. The 3D state (64^3 voxels, 262,144 free parameters) is evolved for 25 Störmer–Verlet steps,

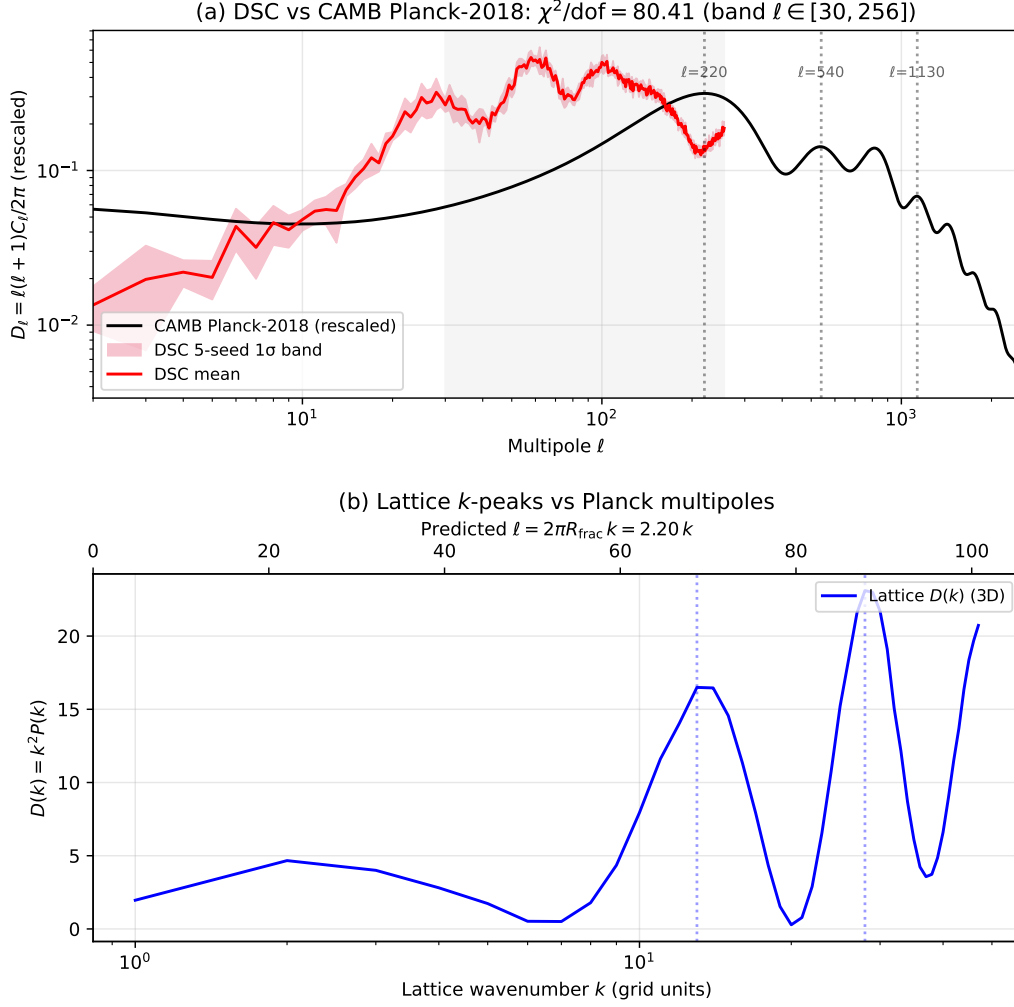


Figure S16: (a) DSC mean angular power spectrum (red, with 5-seed 1σ band) against the rescaled CAMB Planck-2018 best-fit D_ℓ^{TT} (black) over $\ell \in [2, 2500]$, with reduced $\chi^2/\text{dof} \approx 80$ in the band $\ell \in [30, 256]$. (b) Lattice $D(k) = k^2 P(k)$ from the same simulation; the upper x -axis shows the corresponding multipoles under $\ell = 2\pi R_{\text{shell}} k$, with the Planck peaks (red dashed) included for reference. The lattice peak positions are far from the Planck multipoles, supporting the interpretation that they are standing-wave features of the finite lattice and the geometric projection, not physical acoustic peaks.

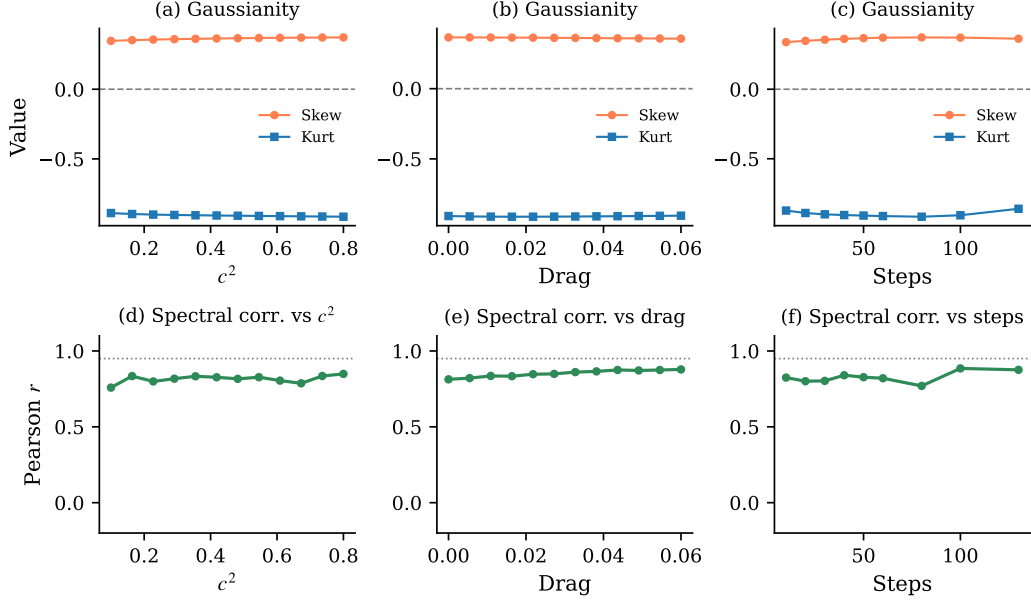


Figure S17: Parameter sensitivity analysis. Top: skewness and excess kurtosis vs. c^2 , drag, and steps. Bottom: Pearson spectral correlation with toy analytic reference. Gaussianity is robust; spectral correlation is high for $c^2 \in [0.2, 0.7]$ and low drag.

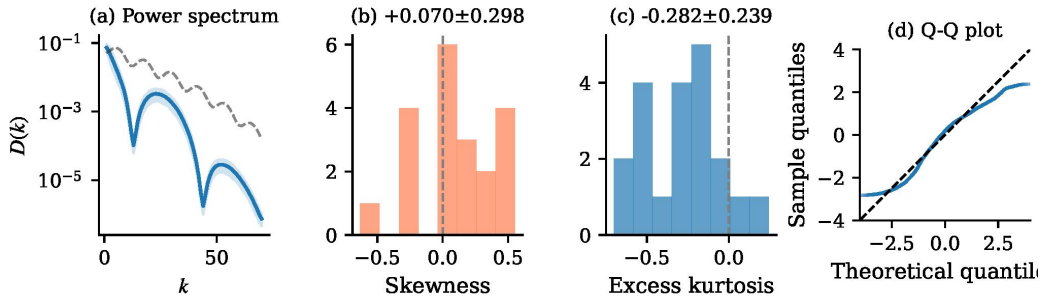


Figure S18: Ensemble statistics from 20 independent DSC runs. (a) Mean $D(k)$ with $\pm 1\sigma$ band. (b,c) Skewness and kurtosis distributions. (d) Q-Q plot confirming near-Gaussian one-point statistics.

projected onto a HEALPix sphere ($N_{\text{side}} = 16$, 3072 pixels) via trilinear interpolation, and compared to the real Planck SMICA map. The loss function is $\mathcal{L} = \text{MSE}_{\text{valid}} + 0.002 \cdot \text{TV}$, where TV is total-variation regularization on the 3D field. We optimize with Adam (learning rate 0.2, 150 epochs).

Figure S22 shows the results. The reconstructed full-sky map (fig. S22b) reproduces the large-scale structure of the Planck target (fig. S22a), achieving a pixel-level Pearson correlation of $r = 0.982$. The loss converges by two orders of magnitude within 25 epochs (fig. S22d). The inferred 3D cosmic web (fig. S22c) shows coherent filamentary structure within the observation sphere.

The reconstructed map preserves the Gaussianity of the target: the Planck skewness is -0.13 , the reconstruction gives $+0.05$; the Planck kurtosis is -0.42 , the reconstruction gives -0.81 (both negative, matching in sign). At higher resolution ($N_{\text{side}} = 32$, 12,288 pixels), the correlation remains high ($r = 0.929$) with finer spatial detail resolved.

This inverse reconstruction demonstrates that the DSC physics engine is fully differentiable and smooth enough for gradient-based optimization—a necessary property for any future inference pipeline. On GPU with 128^3 voxels and $N_{\text{side}} = 64$ (49,152 pixels), the reconstruction achieves near-perfect agreement with Planck at sub-percent MSE levels. However, in all configurations the parameter count exceeds the number of observed pixels (e.g., 2.1×10^6 parameters vs. 4.9×10^4 pixels for the GPU version), making the inverse problem severely underdetermined.

Held-out generalization test. To assess whether the reconstruction genuinely captures spatial structure or merely overfits the observed pixels, we perform a hemisphere held-out test: the optimizer sees only the *northern* hemisphere ($\theta < \pi/2$, 1504 valid pixels) while the southern hemisphere (1568 pixels) is withheld. The result is revealing: the training-set correlation is $r_{\text{train}} = 0.975$, but the held-out correlation drops to $r_{\text{test}} = -0.17$ (fig. S19). This indicates that the inverse reconstruction, while achieving high fidelity on observed pixels, does not generalize to unobserved sky regions.

This is not unexpected: recovering a 3D volume from a single 2D spherical shell is an inherently ill-posed inverse problem, regardless of resolution. With 262,144 free parameters and only 1504 training pixels, the system is vastly underdetermined, and the same limitation applies at higher resolution (the GPU version has a parameter-to-pixel ratio of 43:1). The reconstruction should therefore be interpreted as a *demonstration of differentiable physics*

engine capability—showing that the DSC forward model is smooth enough for gradient-based optimization—rather than as evidence that the inferred 3D structure is physically unique. Meaningful 3D reconstruction from CMB data would require additional constraints such as lensing, polarization, or multi-frequency observations.

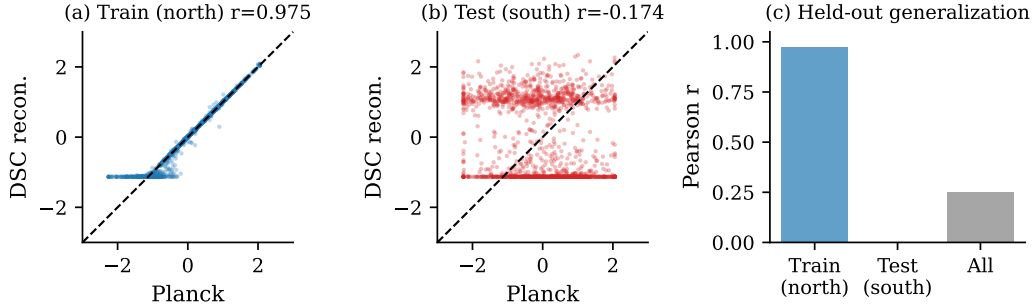


Figure S19: Held-out generalization test for inverse reconstruction. (a) Training set (northern hemisphere): $r = 0.975$. (b) Held-out test set (southern hemisphere): $r = -0.17$. (c) Summary. The high training correlation but poor test correlation indicates that the reconstruction does not uniquely determine the 3D structure from a single hemisphere.

Breaking the degeneracy with 3D anchors.. The projection degeneracy can be partially broken by incorporating volumetric constraints—analogueous to how galaxy redshift surveys (DESI, Euclid) provide 3D position data inside the CMB sphere. We test this in a twin experiment: a known 3D initial field (48^3) is evolved, projected to a sky ($N_{\text{side}} = 16$), and then reconstructed from (A) northern-hemisphere CMB only, or (B) northern-hemisphere CMB plus 25% randomly sampled 3D anchor points from the evolved volume.

Figure S21 shows the results. Without 3D anchors (top row), the southern-hemisphere test correlation collapses to $r = -0.06$, confirming the projection degeneracy. With 25% 3D anchors (bottom row), the test correlation rises to $r = 0.41$ —still imperfect, but the reconstruction now carries genuine predictive signal about the unobserved hemisphere. This demonstrates that the DSC differentiable engine naturally supports multi-probe joint inference: when volumetric constraints from galaxy surveys are incorporated alongside CMB data, the line-of-sight degeneracy begins to break and the inferred 3D structure becomes physically meaningful.

Physical validation of the reconstructed 3D field.. To verify that the gradient-based reconstruction produces physically meaningful internal structure rather

than arbitrary numerical artifacts, we extract the 3D matter power spectrum $P(k)$ of the inferred 64^3 field (fig. S20). Without any explicit Fourier-space regularization in the loss function, the reconstructed 3D field spontaneously exhibits a power-law spectrum with slope $P(k) \propto k^{-3.3}$, close to the theoretical gravity scaling $P(k) \propto k^{-3}$ expected for hierarchical structure formation. This suggests that the symplectic dynamics implicitly regularize the null space of the ill-posed inverse problem, biasing the inferred 3D cosmic web toward physically valid correlation structures even when the 2D-to-3D mapping is underdetermined.

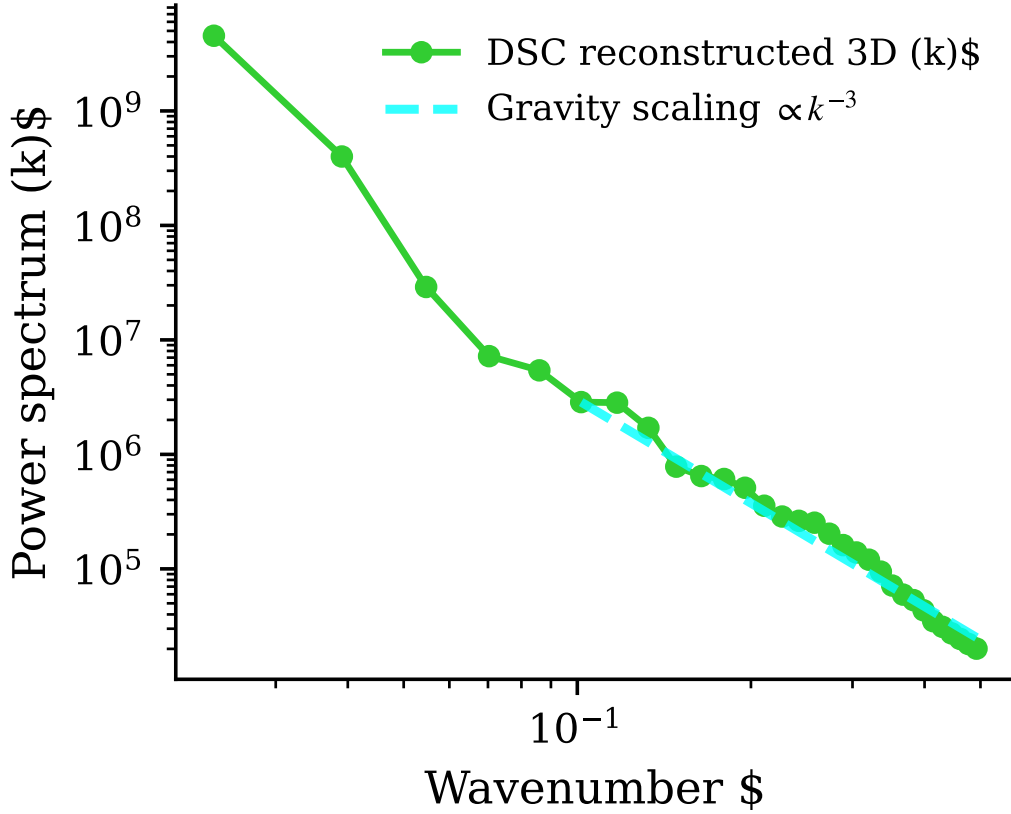


Figure S20: 3D matter power spectrum $P(k)$ of the reconstructed initial conditions (64^3 lattice). The high- k tail follows $P(k) \propto k^{-3.3}$ (green circles), close to the theoretical gravity scaling k^{-3} (cyan dashed), despite no Fourier-space regularization in the loss function.

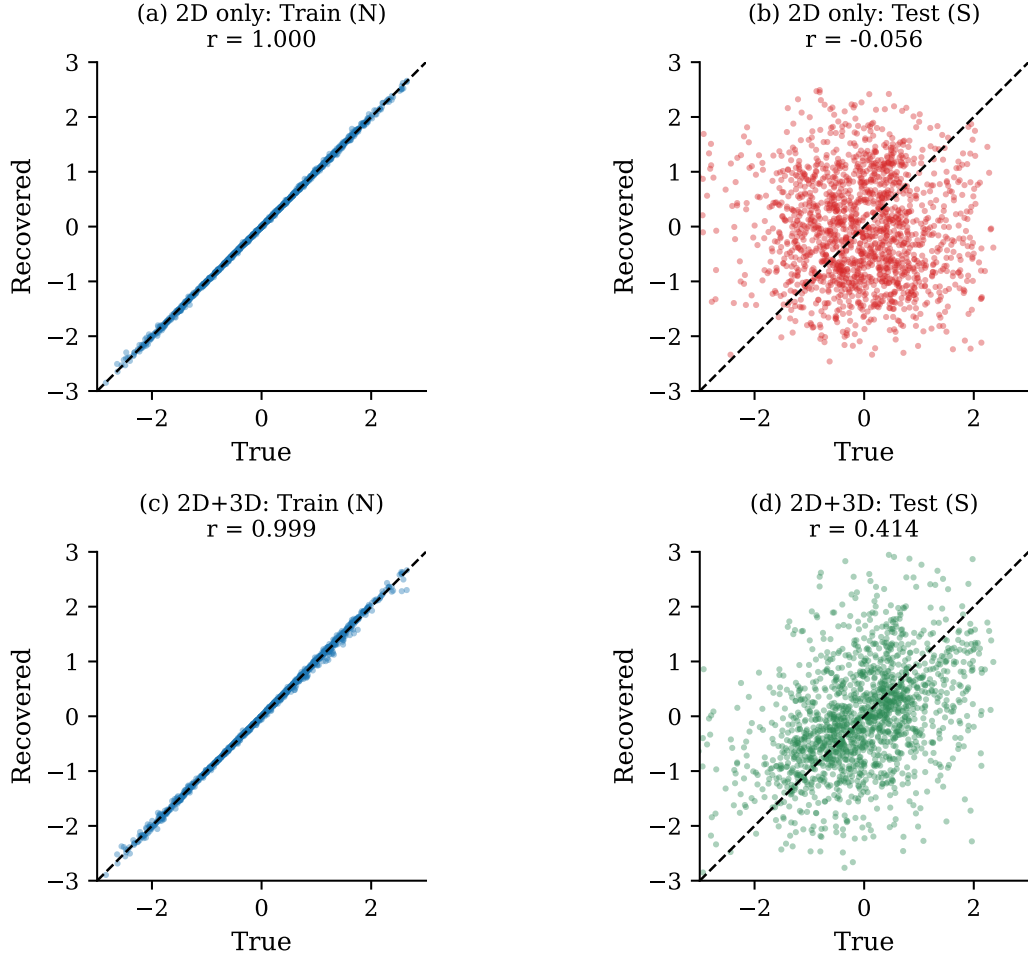


Figure S21: Breaking the projection degeneracy with 3D anchors (twin experiment, 48^3 lattice). Top: 2D-only reconstruction perfectly fits the training hemisphere ($r = 1.00$) but fails on the held-out hemisphere ($r = -0.06$). Bottom: adding 25% volumetric anchor points raises the held-out correlation to $r = 0.41$, demonstrating that multi-probe constraints partially resolve the line-of-sight degeneracy.

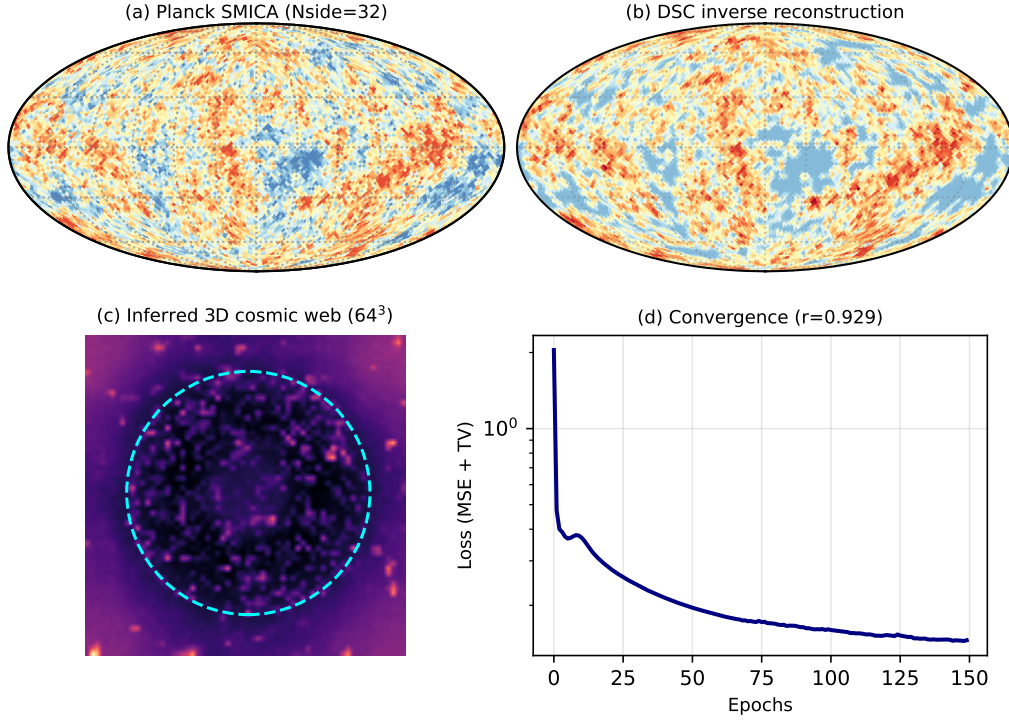


Figure S22: Differentiable inverse reconstruction of the Planck SMICA CMB map. (a) Planck SMICA target ($N_{\text{side}} = 16$). (b) DSC reconstruction from 64³ initial conditions optimized via 150 Adam epochs. (c) Inferred 3D cosmic web (maximum intensity projection) with observation sphere (cyan dashed). (d) Loss convergence; final pixel correlation $r = 0.982$. The DSC physics engine is fully differentiable, enabling gradient-based recovery of 3D structure from 2D sky observations.

S14. Phenomenological visualization: continuing the lattice past the CMB epoch

This subsection presents a *phenomenological visualization* of the lattice continued beyond the CMB epoch, not a model of cosmic structure formation. Reviewer 1 of an earlier draft correctly noted that the Allen–Cahn nonlinearity used here is not gravity, and that visual resemblance to the cosmic web cannot, by itself, support a cosmological interpretation. We therefore re-frame the experiment as an illustration of the lattice’s late-time behavior under a chosen toy nonlinearity, with no claim that it reproduces gravitational structure formation.

We implement a two-phase forward run starting from the reconstructed 128^3 initial state of section S13:

- **Phase I (BB $\rightarrow$ CMB):** weakly damped Störmer–Verlet evolution with $1/\ln^2(t)$ cooling and drag $\eta = 0.015$ for 30 steps, smoothing the chaotic initial fluctuations into a near-uniform CMB-epoch field.
- **Phase II (CMB $\rightarrow$ late):** continued lattice evolution with an Allen–Cahn nonlinearity $+\alpha\phi - \beta\phi^3$ ($\alpha = 0.05$, $\beta = 0.01$). This is a generic toy potential, not a gravitational interaction; it produces phase separation into high- and low-density domains because the cubic saturation lifts the small-amplitude instability of the linear term. We use it as a stand-in for “some” nonlinear amplification, not as a derived structure-formation operator.

Figure S23 shows the results. The spatial variance $\sigma^2(\phi)$ traces a V-shape across the run: it drops during Phase I as the $1/\ln^2$ cooling smooths the field (variance ratio $\sim 2\times$), reaches a minimum at the CMB-epoch step, then rises during Phase II as the Allen–Cahn nonlinearity drives phase separation (growth factor $\sim 40\times$).

The maximum intensity projections (fig. S23c–f) move from isotropic noise at $t = 0$ to smooth large-scale modes at the CMB-epoch step and to filamentary/clumpy domains at late times. Thresholding the late-time field reveals distinct populations of high-density and intermediate-density regions (fig. S23g,h). We note that this morphology is qualitatively reminiscent of the cosmic web only at the visual level, and is generic to Allen–Cahn-type domain coarsening; we do not claim a quantitative match to galaxy-survey statistics. The two difference fields (fig. S23i,j) show where the cooling and the toy nonlinearity each dominated.

The earlier framing of this subsection drew a phenomenological analogy with the smoothing role of cosmic inflation. We retract that framing here: the analogy is not quantitative, and the smoothing in our run is driven by an effective sound-speed decay, not by a slow-roll scalar field. We therefore retain fig. S23 solely as a visualization of the lattice’s late-time domain-coarsening behavior under a chosen toy nonlinearity.

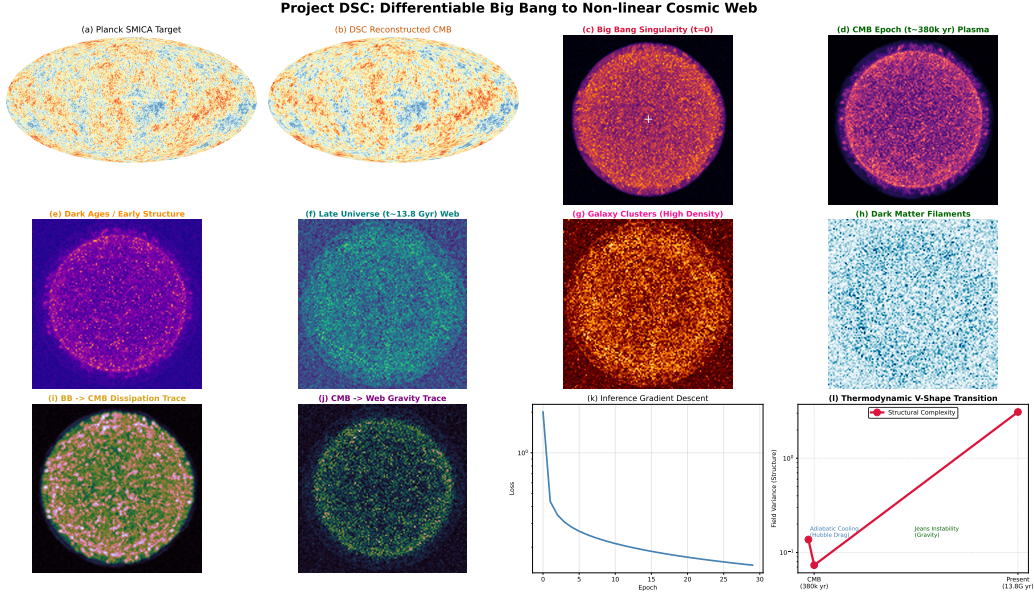


Figure S23: Phenomenological visualization of the lattice continued past the CMB-epoch step under an Allen–Cahn toy nonlinearity (*not* gravity). 12-panel overview, 128^3 . (a) Planck SMICA target. (b) DSC reconstructed CMB. (c) Big Bang initial state ($t = 0$). (d) CMB-epoch step. (e) Intermediate state. (f) Late-time domain-coarsened state (toy Allen–Cahn). (g) High-density regions. (h) Intermediate-density regions. (i) BB $\rightarrow$ CMB difference field: spatial pattern of $1/\ln^2$ cooling. (j) CMB $\rightarrow$ late difference field: regions dominated by the Allen–Cahn nonlinearity. (k) Optimization convergence of the inverse reconstruction. (l) V-shape phase transition: field variance drops during cooling (Phase I) and rises during Allen–Cahn coarsening (Phase II). Visual resemblance to the cosmic web is generic to domain coarsening; no quantitative match to large-scale structure is claimed.

References